

Technology Stack

Date	29 June 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Define Your Technology Stack:

The OptiCrop: Smart Agricultural Production Optimization Engine is developed using a multi-layer technology stack. The frontend uses HTML, CSS, and JavaScript to provide an easy-to-use interface for farmers. The backend is built using Python and Flask to process user inputs and generate crop recommendations. Machine learning libraries such as Scikit-learn, Pandas, and NumPy are used for data analysis and crop prediction. The project uses agricultural datasets stored in CSV/SQLite format and is managed using Git and GitHub for version control and collaboration. This technology stack ensures that the application is scalable, maintainable, and efficient in providing accurate crop recommendations.

TECHNOLOGY STACK DETAILS:

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side(e.g., React, Angular, HTML/CSS)	HTML, CSS, JavaScript	Used to build a simple, responsive, and user-friendly interface where farmers can enter soil parameters and view crop recommendations.

2	Backend / Server-Side (e.g., Node.js, Python/Django, Java)	Python, Flask	Flask handles user requests and connects the frontend with the machine learning model for fast processing and prediction.
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3	Database / Data Storage (e.g., MySQL, MongoDB, PostgreSQL)	CSV Agricultural Dataset and SQLite	Stores crop and soil data efficiently and is easy to manage for machine learning analysis and predictions.
4	Cloud / Hosting / Deployment (e.g., AWS, Azure, Heroku, Docker)	GitHub and Render (Future Deployment)	GitHub is used for project hosting and version management, while Render can be used for easy and costeffective deployment.
5	Version Control & CI/CD (e.g., GitHub, GitLab, Jenkins)	Git and GitHub	Enables team collaboration, tracks code changes, and maintains backup versions of the project.

6	Third-Party APIs / Other Tools (e.g., Stripe, Twilio, Google Maps)	• ☐ Scikit-learn • ☐ Pandas • ☐ NumPy • ☐ Matplotlib • ☐ Seaborn • ☐ Jupyter Notebook • ☐ Visual Studio Code	• ☐ Scikit-learn: Builds and trains the crop recommendation model. • ☐ Pandas: Handles and processes agricultural datasets. • ☐ NumPy: Performs numerical computations. • ☐ Matplotlib and Seaborn: Create charts and visualizations for data analysis.
			☐ Jupyter Notebook: Used for machine learning experimentation and analysis. ☐ Visual Studio Code: Used for coding and project development.